

Why Robust Powerline Perching?

A maintenance and inspection UAV can perch on a powerline to recharge and continue its mission. Wind moves both the wire and the vehicle, creating a potential cue for predicting the target.

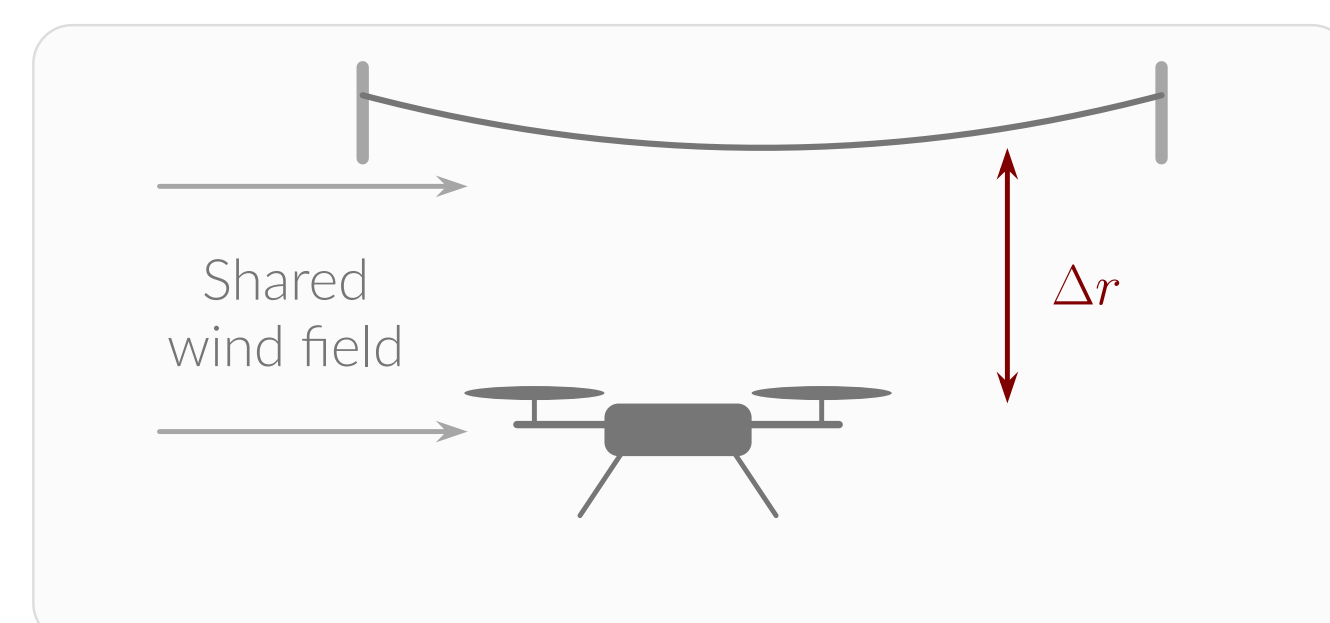


Figure 1. **Core idea:** Vehicle disturbance can help predict wire motion. Its informativeness grows as separation shrinks.

Simulator fidelity must preserve this **shared disturbance structure**, alongside mechanics and contact.

Contributions

- κ links sag and separation to vehicle–wire wind coherence
- Calibrated simulator with tested spatial wind correlation
- Proposed controller comparison to test the benefit of vehicle wind cues

Formalizing Wind–Wire Coupling

For small out-of-plane swing of a shallow span:

$$f_1 = \frac{1}{2} \sqrt{\frac{g}{8d}} \approx 0.554 d^{-1/2} \text{ Hz}$$

Coherence at f_1 , for vertical separation Δr :

$$\kappa = \frac{C f_1 \Delta r}{U}, \quad \sqrt{\text{Coh}(f_1, \Delta r)} = e^{-\kappa}$$

d : sag (m); U : mean wind speed; C : coherence-decay coefficient.

Turbulent Winds and Spatial Coherence

The simulator uses spatially correlated Dryden turbulence, matched to Davenport's root-coherence at f_1 .

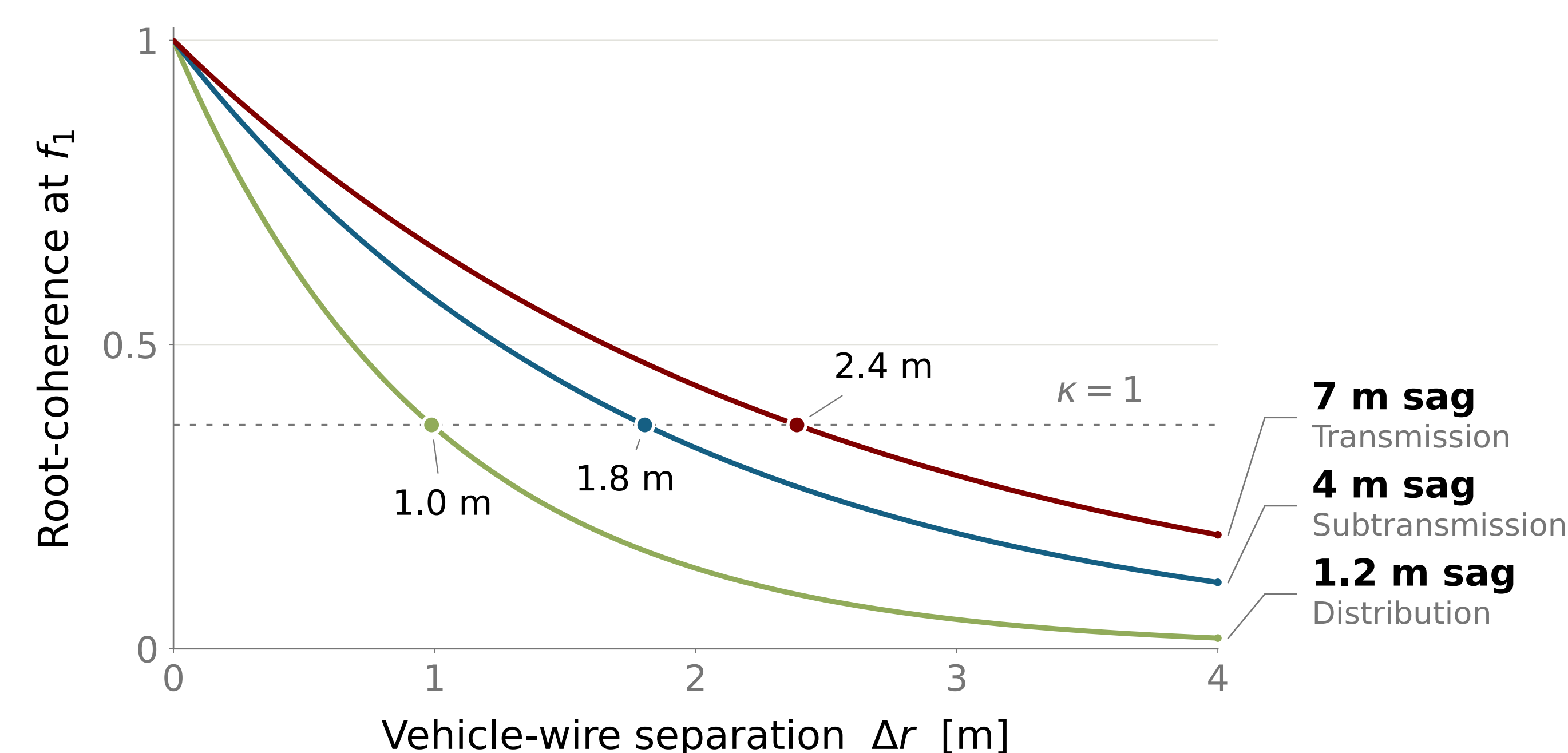


Figure 2. Analytical curves for representative sags: vertical separation, $U = 5 \text{ m/s}$, $C = 10$; dots mark $\kappa = 1$.

Simulator Validation

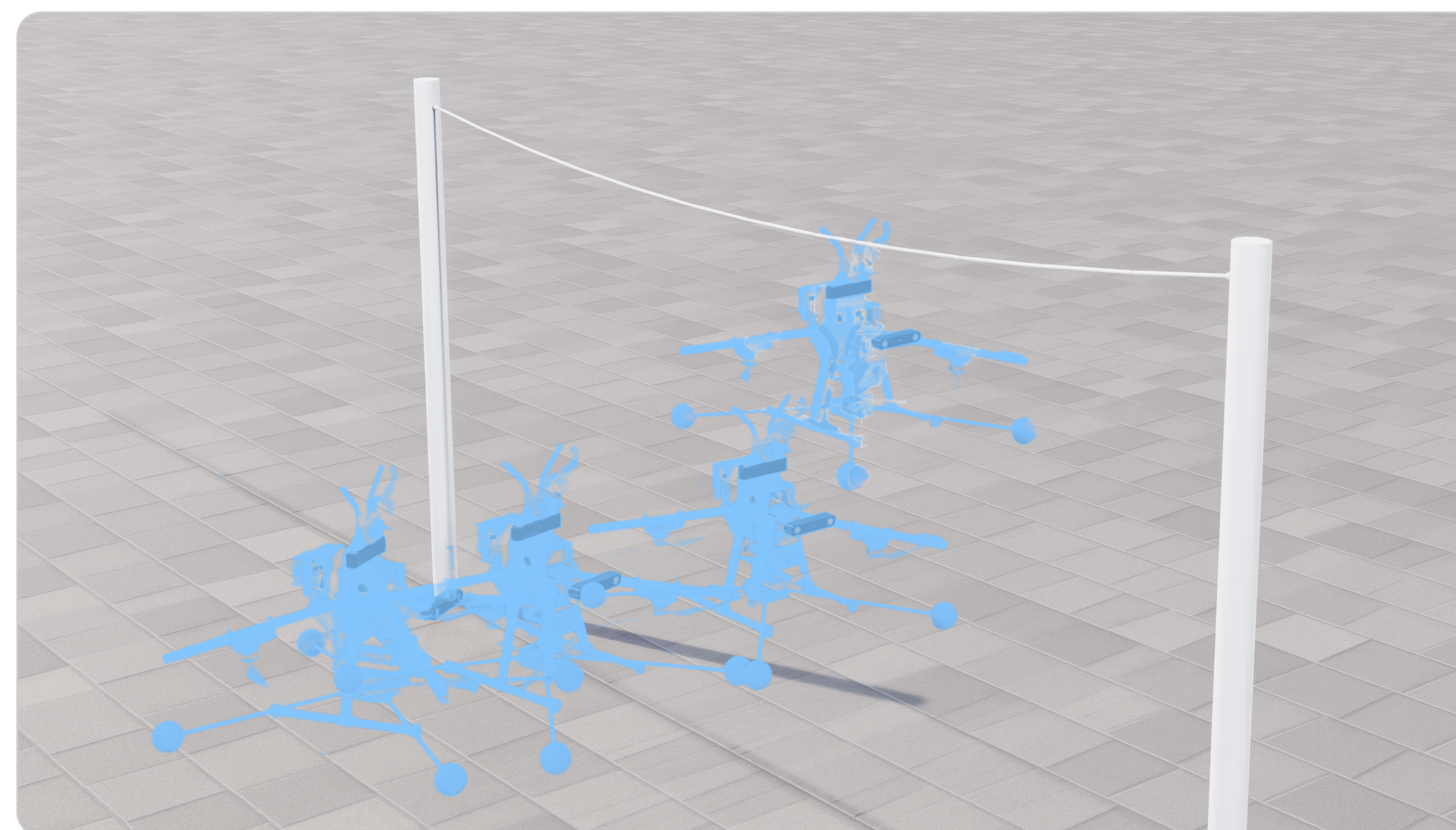


Figure 3. Visual-servoing baseline: the drone's path during approach and ascent to perch on the conductor.

Built in NVIDIA Isaac Sim with inertia from CAD and actuation calibrated to flight logs. Conductor tension emerges from span geometry.

Simulator Fidelity

Check	Reference	Relative error
Static wind tilt	Analytical tilt at 5 m/s	0.29%
Frequency ratio	Analytical; 2.5× sag range	0.33%
Airframe mass	CAD vs. measurement	0.98%

End-to-end check: simulated latch using a supplied wire position.

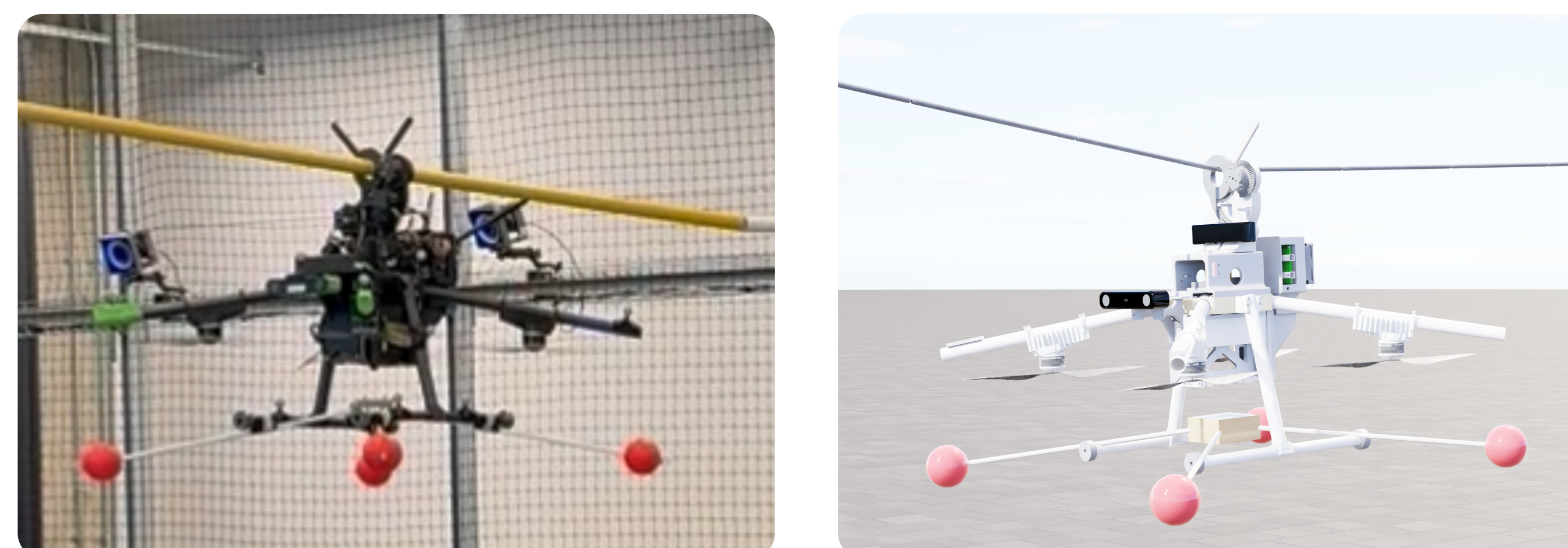
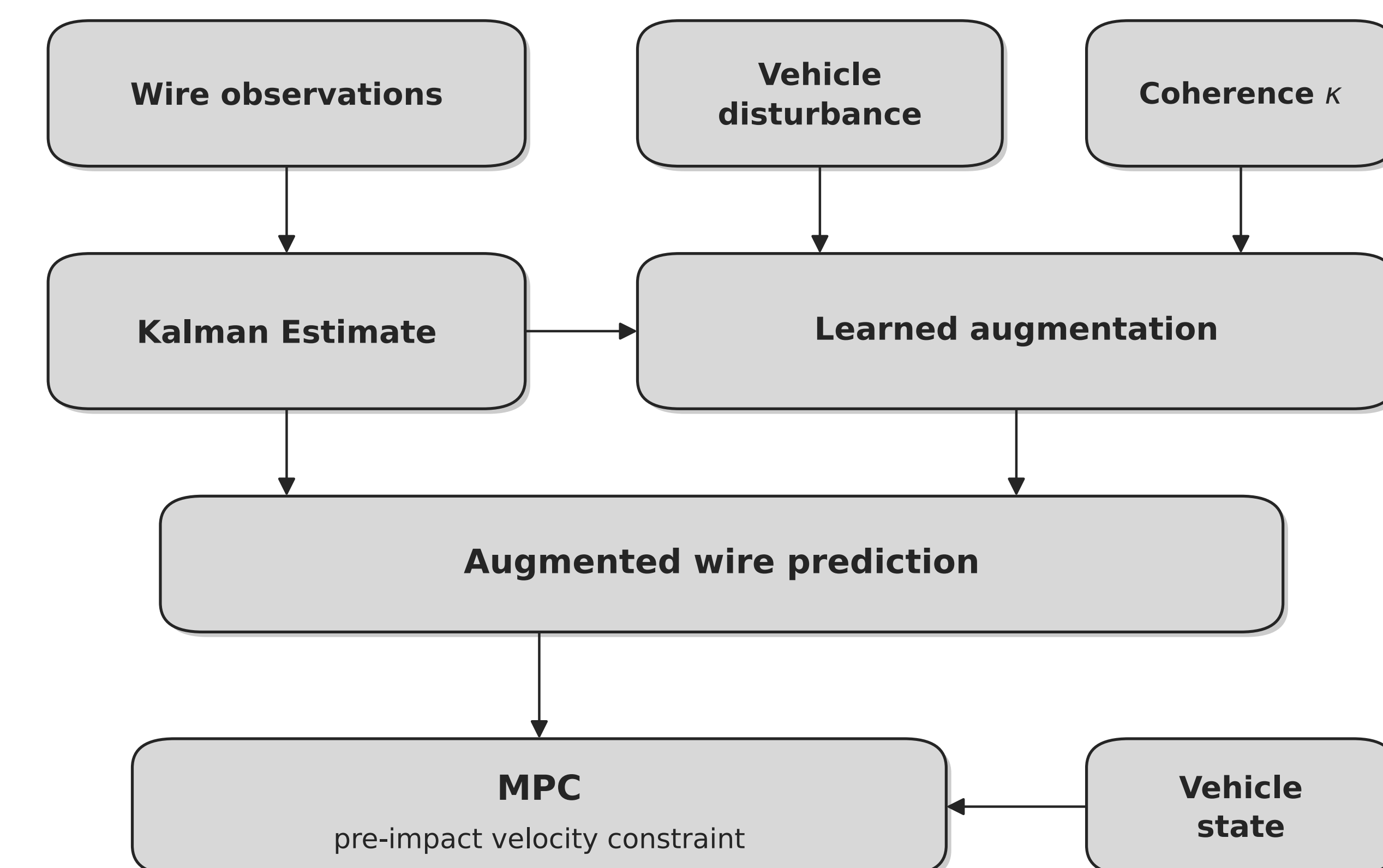


Figure 4. Contact-triggered latch, hardware (left) and simulation (right). Stills illustrate the latch mechanism.

Proposed Wind-Informed Control



Identifying the Benefit of Wind Cues

We will compare four approaches to the same perching task, assessed against the same pre-impact relative-velocity limit. Use the same MPC in (2) and (3).

- 1 Visual servoing**
Regulate on the live wire estimate.
- 1.5 Predictive servoing**
Add the oscillator model to the same regulator.
- 2 MPC**
Add a planning horizon with a pre-impact velocity constraint.
- 3 MPC + learned augmentation**
Augment the wire predictor with the vehicle's own disturbance.

Key Experimental Risk

Stronger wind coherence may not translate into better wire predictions: vehicle disturbance may add little beyond wire observations, or the learned augmentation may fail to exploit it. Compare (3) against (2) on matched wind realizations to test its benefit.